

# pylhe: A Lightweight Python interface to Les Houches Event files

Alexander Puck Neuwirth<sup>1,2</sup>, Matthew Feickert<sup>3</sup>, Lukas Heinrich<sup>4</sup>, and Eduardo Rodrigues<sup>5</sup>

<sup>1</sup> University of Milan Bicocca <sup>2</sup> INFN Milan Bicocca <sup>3</sup> University of Wisconsin-Madison <sup>4</sup> Technical University of Munich <sup>5</sup> University of Liverpool ¶ Corresponding author

DOI: [10.21105/joss.11128](https://doi.org/10.21105/joss.11128)

## Software

- [Review](#) ↗
- [Repository](#) ↗
- [Archive](#) ↗

Editor: [Kyle Niemeyer](#) ↗

## Reviewers:

- [@BFuks](#)
- [@Moelf](#)
- [@gavinsdavies](#)

Submitted: 15 December 2025

Published: 06 October 2026

## License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

## Summary

pylhe is a lightweight Python library that provides a simple and efficient interface for reading and writing Les Houches Event (LHE) files, a standard format used by Monte Carlo event generators in high-energy physics ([Alwall & others, 2007](#)). The library enables memory-efficient streaming of events from XML format (`.lhe`, compressed `.lhe.gz`) and HDF5 format (`.h5`, `.hdf5`, `lheh5`) files through a pythonic iterator interface, allowing researchers to process arbitrarily large event files without loading all events into memory simultaneously. It is a standalone, event-generator-independent package, installable from PyPI and Conda. pylhe, part of the Scikit-HEP community project ([Rodrigues & others, 2020](#); [The Scikit-HEP Project, 2016](#)), is well maintained and integrated within the broader scientific Python ecosystem.

Historically, the first standards for event representation in high-energy physics were the HEPEVT and HEPRUP common blocks ([Boos & others, 2001](#)), which provided a Fortran-based structure for storing event information. As the complexity of Monte Carlo event generators increased, the need for a more flexible and extensible format led to the development of the LHE file format. LHE introduced an XML structure that allowed for better organization of event data and facilitated interoperability between different tools. Typically, LHE files are used to describe parton-level events generated by matrix element generators that are passed to parton shower and hadronization programs producing HepMC files ([Buckley et al., 2021](#); [Dembinski et al., 2022](#); [Dobbs & Hansen, 2001](#)), or directly to analysis frameworks such as Rivet ([Bierlich et al., 2024](#); [Bierlich & others, 2020](#); [Buckley et al., 2013](#)). In contrast to LHE files, HepMC stores fewer details about the particles, but also contains many more particles per event. Both files are plain text-based formats, making them human-readable but potentially large. Consequently, compression using the `gzip` library is common practice for both formats.

Since pylhe version 2.0.0, the library also supports the HDF5-based ([The HDF Group, 1998](#)) LHEH5 format ([Bothmann et al., 2024](#); [Höche et al., 2019](#)), which allows for significant improvements in write performance while keeping file sizes for event storage under control.

The LHE format stores information within `<init>` and multiple `<event>` blocks consisting of whitespace-separated values designed historically for straightforward parsing in Fortran. Further details can be found in the original definition of the Les Houches Event file standard ([Alwall & others, 2007](#)). Following the original publication there were two extensions to the LHE format, version 2.0 in 2009 ([Butterworth & others, 2010](#)) and version 3.0 in 2013 ([Andersen & others, 2014](#)). pylhe completely implements the widely adopted version 3.0, that is the addition of `<scales>`, `<generator>`, and multiple weights via `<initrwgt>`, `<rwgt>`, `<weight>`, `<weights>`, `<wgt>`, and `<weightgroup>` on top of the version 1.0. Features that were introduced in version 2.0 but removed in version 3.0, such as `<pdfinfo>` or `<clustering>`, are not supported.

## Statement of need

The LHE format is used by all major Monte Carlo event generators such as MadGraph ([Alwall et al., 2014](#)), POWHEG-BOX ([Alioli et al., 2010](#); [Frixione et al., 2007](#); [Nason, 2004](#)), Sherpa ([Bothmann & others, 2019](#); [Gleisberg et al., 2009](#)), HERWIG ([Bahr & others, 2008](#); [Bellm & others, 2016, 2020](#); [Bewick & others, 2024](#); [Corcella et al., 2001](#)), Pythia ([Bierlich & others, 2022](#); [Sjostrand et al., 2006, 2008](#); [Sjöstrand et al., 2015](#)), Whizard ([Kilian et al., 2011](#); [Moretti et al., 2001](#)). While interfaces for C/C++/Fortran exist in the respective generators, a lightweight, standalone and easy-to-use Python interface equally supporting all these event generators was missing until the inception of pylhe in 2015. Additionally, pylhe can serve as a crucial interface for emerging machine learning applications in particle physics, allowing researchers to efficiently extract event data for training neural networks and other machine learning models used in event classification, anomaly detection, and physics analysis. Past experience and the ever-growing number of publications citing pylhe have shown that the portfolio of users goes from experimentalists to phenomenologists, some of which are themselves developers of event generators.

As a library, pylhe is straightforwardly installable via the standard means, and is maintained within the Scikit-HEP community project that is very well integrated within the broader scientific Python ecosystem.

## State of the field

Unlike the existing C/C++/Fortran interfaces provided by the Monte Carlo event generators, pylhe offers a standalone, pure Python interface that is more accessible and easier to use. When pylhe was first developed, there were no other standalone Python libraries available for reading and writing LHE files. However, a Python parser is distributed as part of MadGraph ([Alwall et al., 2014](#)), while MadAnalysis 5 exposes its C++ LHE reader and writer through a Python interface ([Conte et al., 2013](#)). Nowadays, there are a few other smaller Python libraries with less adoption than pylhe, which provide only read functionality and are no longer actively maintained, such as lhereader ([Albert, 2021](#); [Biswas, 2017](#)). For completeness, it should be mentioned that several LHE libraries exist in other programming languages, such as Go ([go-hep, Binet et al., 2017](#)), Rust ([lhe, Fukuda, 2018](#); [lhef, Maier, 2024b](#); [Weber, 2018](#); [event\\_file\\_reader, Maier, 2024a](#)), Julia ([LHEF.jl, Fischer & Ling, 2023](#)) and Haskell ([lhe.hs, Lawrence, 2019](#)). These provide varying degrees of completeness, but none support the LHEH5 format yet.

## Software design

pylhe allows for easy reading and writing of `.lhe` and `.lhe.gz` files in Python, enabling seamless integration into modern data analysis workflows in high-energy physics. The pythonic event yielding approach allows for memory-efficient processing of arbitrarily large LHE files by streaming events one at a time rather than loading all of them at once into memory. Internally, pylhe uses `xml.etree.ElementTree` to parse the XML structure, since using the `lxml` library did not provide a significant speed up. For LHEH5 files the library uses `h5py` ([Collette, 2013](#)) to read and write the HDF5 format, via the same pythonic data structures, making it easy to convert between the standard XML and newer HDF5 formats.

The library facilitates quick validation of event files through programmatic access to event structure and particle properties, making it straightforward to perform sanity checks on generated events. This can be done for example via the integration with Scikit-HEP's Awkward Array package `awkward` ([Pivarski et al., 2018](#)) through the `to_awkward()` function, which converts LHE events into columnar data structures optimized for vectorized operations and efficient analysis of large datasets.

The library endeavours to support all major Monte Carlo event generators and relies on example XML and HDF5 format files to ensure test coverage.

## Research impact statement

pylhe is regularly used in various research projects and publications within high-energy physics. Notably, it has been cited in Higgs studies (Brehmer et al., 2019; Feuerstake et al., 2025; Stylianou & Weiglein, 2024), in Supersymmetry (SUSY), Beyond the Standard Model (BSM) and dark matter searches (Anisha et al., 2023; Beresford et al., 2024; Beresford & Liu, 2019; Cheung et al., 2024; Kling, 2020; Zhou & Liu, 2022, 2025), and in forward physics studies (Kelly et al., 2022; Kling et al., 2023; Kling & Trojanowski, 2020). It is also employed in methodological studies involving machine learning techniques for event generation and analysis (Brehmer et al., 2020; Kofler et al., 2025).

## AI usage disclosure

Generative AI tools have been used in the development of this software and writing of the manuscript:

- Github's copilot has been used in reviewing pull requests.
- VScode's copilot has been used as an advanced autocomplete.
- ChatGPT has been used to identify the most pythonic solutions in case of ambiguity.

All the results generated by these tools have been reviewed by the authors and are correct to the best of our knowledge.

## Author order

The authors agreed to list Alexander Puck Neuwirth first, with the remaining authors listed alphabetically by family name.

## Acknowledgements

We would additionally like to thank the contributors of pylhe and the Scikit-HEP community for their support.

## References

- Albert, A. (2021). *Lhereader*. GitHub. <https://github.com/AndreasAlbert/lhereader>
- Alioli, S., Nason, P., Oleari, C., & Re, E. (2010). A general framework for implementing NLO calculations in shower Monte Carlo programs: the POWHEG BOX. *JHEP*, 06, 043. [https://doi.org/10.1007/JHEP06\(2010\)043](https://doi.org/10.1007/JHEP06(2010)043)
- Alwall, J., Frederix, R., Frixione, S., Hirschi, V., Maltoni, F., Mattelaer, O., Shao, H.-S., Stelzer, T., Torrielli, P., & Zaro, M. (2014). The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations. *JHEP*, 07, 079. [https://doi.org/10.1007/JHEP07\(2014\)079](https://doi.org/10.1007/JHEP07(2014)079)
- Alwall, J., & others. (2007). A Standard format for Les Houches event files. *Comput. Phys. Commun.*, 176, 300–304. <https://doi.org/10.1016/j.cpc.2006.11.010>
- Andersen, J. R., & others. (2014). *Les Houches 2013: Physics at TeV Colliders: Standard Model Working Group Report*. <https://doi.org/10.48550/arXiv.1405.1067>

- Anisha, Atkinson, O., Bhardwaj, A., Englert, C., Naskar, W., & Stylianou, P. (2023). BSM reach of four-top production at the LHC. *Phys. Rev. D*, 108(3), 035001. <https://doi.org/10.1103/PhysRevD.108.035001>
- Bahr, M., & others. (2008). Herwig++ Physics and Manual. *Eur. Phys. J. C*, 58, 639–707. <https://doi.org/10.1140/epjc/s10052-008-0798-9>
- Bellm, J., & others. (2016). Herwig 7.0/Herwig++ 3.0 release note. *Eur. Phys. J. C*, 76(4), 196. <https://doi.org/10.1140/epjc/s10052-016-4018-8>
- Bellm, J., & others. (2020). Herwig 7.2 release note. *Eur. Phys. J. C*, 80(5), 452. <https://doi.org/10.1140/epjc/s10052-020-8011-x>
- Beresford, L., Clawson, S., & Liu, J. (2024). Strategy to measure tau g-2 via photon fusion in LHC proton collisions. *Phys. Rev. D*, 110(9), 092016. <https://doi.org/10.1103/PhysRevD.110.092016>
- Beresford, L., & Liu, J. (2019). Search Strategy for Sleptons and Dark Matter Using the LHC as a Photon Collider. *Phys. Rev. Lett.*, 123(14), 141801. <https://doi.org/10.1103/PhysRevLett.123.141801>
- Bewick, G., & others. (2024). Herwig 7.3 release note. *Eur. Phys. J. C*, 84(10), 1053. <https://doi.org/10.1140/epjc/s10052-024-13211-9>
- Bierlich, C., Buckley, A., Butterworth, J. M., Gutsche, C., Lonnblad, L., Procter, T., Richardson, P., & Yeh, Y. (2024). Robust independent validation of experiment and theory: Rivet version 4 release note. *SciPost Phys. Codeb.*, 36, 1. <https://doi.org/10.21468/SciPostPhysCodeb.36>
- Bierlich, C., & others. (2020). Robust Independent Validation of Experiment and Theory: Rivet version 3. *SciPost Phys.*, 8, 026. <https://doi.org/10.21468/SciPostPhys.8.2.026>
- Bierlich, C., & others. (2022). A comprehensive guide to the physics and usage of PYTHIA 8.3. *SciPost Phys. Codeb.*, 2022, 8. <https://doi.org/10.21468/SciPostPhysCodeb.8>
- Binet, S., Wieck, B., Blyth, D., Busato, E., Ughetto, M., & Waller, P. (2017). Go-HEP: Libraries for high energy physics analyses in go. *Journal of Open Source Software*, 2(17), 372. <https://doi.org/10.21105/joss.00372>
- Biswas, D. (2017). *Lhereader*. GitHub. <https://github.com/diptaparna/lhereader>
- Boos, E., & others. (2001, September). Generic User Process Interface for Event Generators. *2nd Les Houches Workshop on Physics at TeV Colliders*. <https://arxiv.org/abs/hep-ph/0109068>
- Bothmann, E., Childers, T., Gutsche, C., Höche, S., Hovland, P., Isaacson, J., Knobbe, M., & Latham, R. (2024). Efficient precision simulation of processes with many-jet final states at the LHC. *Phys. Rev. D*, 109(1), 014013. <https://doi.org/10.1103/PhysRevD.109.014013>
- Bothmann, E., & others. (2019). Event Generation with Sherpa 2.2. *SciPost Phys.*, 7(3), 034. <https://doi.org/10.21468/SciPostPhys.7.3.034>
- Brehmer, J., Dawson, S., Homiller, S., Kling, F., & Plehn, T. (2019). Benchmarking simplified template cross sections in  $WH$  production. *JHEP*, 11, 034. [https://doi.org/10.1007/JHEP11\(2019\)034](https://doi.org/10.1007/JHEP11(2019)034)
- Brehmer, J., Kling, F., Espejo, I., & Cranmer, K. (2020). MadMiner: Machine learning-based inference for particle physics. *Comput. Softw. Big Sci.*, 4(1), 3. <https://doi.org/10.1007/s41781-020-0035-2>
- Buckley, A., Butterworth, J., Grellscheid, D., Hoeth, H., Lonnblad, L., Monk, J., Schulz, H., & Siebert, F. (2013). Rivet user manual. *Comput. Phys. Commun.*, 184, 2803–2819. <https://doi.org/10.1016/j.cpc.2013.05.021>

- Buckley, A., Ilten, P., Konstantinov, D., Lönnblad, L., Monk, J., Pokorski, W., Przedzinski, T., & Verbytskyi, A. (2021). The HepMC3 event record library for Monte Carlo event generators. *Comput. Phys. Commun.*, 260, 107310. <https://doi.org/10.1016/j.cpc.2020.107310>
- Butterworth, J. M., & others. (2010, March). THE TOOLS AND MONTE CARLO WORKING GROUP Summary Report from the Les Houches 2009 Workshop on TeV Colliders. *6th Les Houches Workshop on Physics at TeV Colliders*. <https://doi.org/10.48550/arXiv.1003.1643>
- Cheung, K., Kim, Y., Kwon, Y., Ouseph, C. J., Soffer, A., & Wang, Z. S. (2024). Probing dark photons from a light scalar at Belle II. *JHEP*, 05, 094. [https://doi.org/10.1007/JHEP05\(2024\)094](https://doi.org/10.1007/JHEP05(2024)094)
- Collette, A. (2013). *Python and HDF5: Unlocking scientific data* (p. 152). O'Reilly Media, Inc. ISBN: 9781491945001
- Conte, E., Fuks, B., & Serret, G. (2013). MadAnalysis 5, A User-Friendly Framework for Collider Phenomenology. *Comput. Phys. Commun.*, 184, 222–256. <https://doi.org/10.1016/j.cpc.2012.09.009>
- Corcella, G., Knowles, I. G., Marchesini, G., Moretti, S., Odagiri, K., Richardson, P., Seymour, M. H., & Webber, B. R. (2001). HERWIG 6: An Event generator for hadron emission reactions with interfering gluons (including supersymmetric processes). *JHEP*, 01, 010. <https://doi.org/10.1088/1126-6708/2001/01/010>
- Dembinski, H., Rodrigues, E., & Fedynitch, A. (2022). *Scikit-hep/pyhepmc: v2.1.1* (Version v2.1.1). Zenodo. <https://doi.org/10.5281/zenodo.7013498>
- Dobbs, M., & Hansen, J. B. (2001). The HepMC C++ Monte Carlo event record for High Energy Physics. *Comput. Phys. Commun.*, 134, 41–46. [https://doi.org/10.1016/S0010-4655\(00\)00189-2](https://doi.org/10.1016/S0010-4655(00)00189-2)
- Feuerstake, F., Fuchs, E., Robens, T., & Winterbottom, D. (2025). Interference effects in resonant di-Higgs production at the LHC in the Higgs singlet extension. *JHEP*, 04, 094. [https://doi.org/10.1007/JHEP04\(2025\)094](https://doi.org/10.1007/JHEP04(2025)094)
- Fischer, K., & Ling, J. (2023). *LHEF.jl*. GitHub. <https://github.com/JuliaHEP/LHEF.jl>
- Frixione, S., Nason, P., & Oleari, C. (2007). Matching NLO QCD computations with Parton Shower simulations: the POWHEG method. *JHEP*, 11, 070. <https://doi.org/10.1088/1126-6708/2007/11/070>
- Fukuda, H. (2018). *Lhe-rs*. GitHub. <https://github.com/hajifkd/lhe-rs>
- Gleisberg, T., Hoeche, Stefan., Krauss, F., Schonherr, M., Schumann, S., Siegert, F., & Winter, J. (2009). Event generation with SHERPA 1.1. *JHEP*, 02, 007. <https://doi.org/10.1088/1126-6708/2009/02/007>
- Höche, S., Prestel, S., & Schulz, H. (2019). Simulation of Vector Boson Plus Many Jet Final States at the High Luminosity LHC. *Phys. Rev. D*, 100(1), 014024. <https://doi.org/10.1103/PhysRevD.100.014024>
- Kelly, K. J., Kling, F., Tuckler, D., & Zhang, Y. (2022). Probing neutrino-portal dark matter at the Forward Physics Facility. *Phys. Rev. D*, 105(7), 075026. <https://doi.org/10.1103/PhysRevD.105.075026>
- Kilian, W., Ohl, T., & Reuter, J. (2011). WHIZARD: Simulating Multi-Particle Processes at LHC and ILC. *Eur. Phys. J. C*, 71, 1742. <https://doi.org/10.1140/epjc/s10052-011-1742-y>
- Kling, F. (2020). Probing light gauge bosons in tau neutrino experiments. *Phys. Rev. D*, 102(1), 015007. <https://doi.org/10.1103/PhysRevD.102.015007>
- Kling, F., Kuo, J.-L., Trojanowski, S., & Tsai, Y.-D. (2023). FLArE up dark sectors with EM form factors at the LHC forward physics facility. *Nucl. Phys. B*, 987, 116103.

- <https://doi.org/10.1016/j.nuclphysb.2023.116103>
- Kling, F., & Trojanowski, S. (2020). Looking forward to test the KOTO anomaly with FASER. *Phys. Rev. D*, 102(1), 015032. <https://doi.org/10.1103/PhysRevD.102.015032>
- Kofler, A., Stimper, V., Mikhasenko, M., Kagan, M., & Heinrich, L. (2025). Flow annealed importance sampling bootstrap meets differentiable particle physics. *Mach. Learn. Sci. Tech.*, 6(2), 025061. <https://doi.org/10.1088/2632-2153/addbc1>
- Lawrence, S. (2019). *Lhe.hs*. GitHub. <https://github.com/bytbox/lhe.hs>
- Maier, A. (2024a). *Event-file-reader*. GitHub. <https://github.com/a-maier/event-file-reader>
- Maier, A. (2024b). *Lhef*. GitHub. <https://github.com/a-maier/lhef>
- Moretti, M., Ohl, T., & Reuter, J. (2001). *O'Mega: An Optimizing matrix element generator*. 1981–2009. <https://doi.org/10.48550/arXiv.hep-ph/0102195>
- Nason, P. (2004). A New method for combining NLO QCD with shower Monte Carlo algorithms. *JHEP*, 11, 040. <https://doi.org/10.1088/1126-6708/2004/11/040>
- Pivarski, J., Osborne, I., Ifrim, I., Schreiner, H., Hollands, A., Biswas, A., Das, P., Roy Choudhury, S., Smith, N., Goyal, M., Fackeldey, P., & Krommydas, I. (2018). *Awkward Array*. <https://doi.org/10.5281/zenodo.4341376>
- Rodrigues, E., & others. (2020). The Scikit HEP Project – overview and prospects. *EPJ Web Conf.*, 245, 06028. <https://doi.org/10.1051/epjconf/202024506028>
- Sjostrand, T., Mrenna, S., & Skands, P. Z. (2006). PYTHIA 6.4 Physics and Manual. *JHEP*, 05, 026. <https://doi.org/10.1088/1126-6708/2006/05/026>
- Sjostrand, T., Mrenna, S., & Skands, P. Z. (2008). A Brief Introduction to PYTHIA 8.1. *Comput. Phys. Commun.*, 178, 852–867. <https://doi.org/10.1016/j.cpc.2008.01.036>
- Sjöstrand, T., Ask, S., Christiansen, J. R., Corke, R., Desai, N., Ilten, P., Mrenna, S., Prestel, S., Rasmussen, C. O., & Skands, P. Z. (2015). An introduction to PYTHIA 8.2. *Comput. Phys. Commun.*, 191, 159–177. <https://doi.org/10.1016/j.cpc.2015.01.024>
- Stylianou, P., & Weiglein, G. (2024). Constraints on the trilinear and quartic Higgs couplings from triple Higgs production at the LHC and beyond. *Eur. Phys. J. C*, 84(4), 366. <https://doi.org/10.1140/epjc/s10052-024-12722-9>
- The HDF Group. (1998). *HDF5 – hierarchical data format version 5*. <https://www.hdfgroup.org/HDF5/>
- The Scikit-HEP Project*. (2016). <https://scikit-hep.org/>
- Weber, T. (2018). *Lhef-rs*. GitHub. <https://github.com/tweber12/lhef-rs>
- Zhou, H., & Liu, N. (2022). Probing compressed higgsinos with forward protons at the LHC. *JHEP*, 10, 092. [https://doi.org/10.1007/JHEP10\(2022\)092](https://doi.org/10.1007/JHEP10(2022)092)
- Zhou, H., & Liu, N. (2025). Search for nearly degenerate higgsinos via photon fusion with the semileptonic channel at the LHC. *Nucl. Phys. B*, 1010, 116752. <https://doi.org/10.1016/j.nuclphysb.2024.116752>